
VTI LEARNING LAB / EXAM QUESTION BANK

Certified Kubernetes Administrator

40 verified practice questions for Kubernetes v1.35

40 questions	CKA exam code
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Community learning material, not an official exam. Only content marked Verified should be used to memorize an answer.

QUESTION 1 / Troubleshooting 30%

A Pod is always in Pending state. Which command should be run first to find the cause?

- A. `kubectl logs pod-name`
- B. `kubectl describe pod pod-name`
- C. `kubectl exec -it pod-name -- sh`
- D. `kubectl delete pod pod-name`

Answer: B - Verified

Pod Pending is often not scheduled or the volume has not been mounted. The Events section of ``kubectl describe`` shows the direct cause.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/debug/debug-application/debug-pods/>

QUESTION 2 / Troubleshooting 30%

The container is CrashLoopBackOff and has restarted. Which command views the log of the failed run?

```
kubectl logs api-7c9d -c api --previous
```

- A. Use the command above
- B. `kubectl get events --watch-only`
- C. `kubectl attach api-7c9d`
- D. `kubectl logs api-7c9d --follow --since=0`

Answer: A - Verified

``--previous`` retrieves the log of the previously terminated container instance; ``-c`` selects the correct container in a multi-container Pod.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/debug/debug-application/debug-running-pod/>

QUESTION 3 / Troubleshooting 30%

A worker node switches to NotReady. Which two tests help localize the kubelet fastest?

- A. `kubectl describe node worker-1`
- B. `journalctl -u kubelet (worker-1)`
- C. `kubectl rollout undo deployment/example`
- D. Delete default namespace

Answer: A, B - Verified

Node conditions and Events indicate signals from the control plane; kubelet journal shows runtime, certificate, cgroup or API connection errors on the node.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/debug/debug-cluster/>

QUESTION 4 / Troubleshooting 30%

After repairing the control plane, kube-apiserver does not start. With kubeadm cluster, which location should be checked first?

- A. /etc/kubernetes/manifests/kube-apiserver.yaml
- B. /var/lib/kube-proxy/config.json
- C. /etc/coredns/Corefile
- D. /opt/kubernetes/deployments.yaml

Answer: A - Verified

kubeadm triển khai các thành phần control plane dưới dạng static Pod. Kubelet theo dõi manifest trong `/etc/kubernetes/manifests` và tạo lại Pod khi file đổi.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/configure-pod-container/static-pod/>

QUESTION 5 / Troubleshooting 30%

Service cannot route to Pods that are Running. Which two tests are most useful?

- A. Compare the Service's selector with the Pod's label
- B. Check the port and targetPort of the Service
- C. Increase CoreDNS replicas instantly
- D. Delete the namespace and then create again

Answer: A, B - Verified

Selector determines which backend appears in EndpointSlice; port/targetPort decides which port of the Pod the traffic is routed to.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/debug/debug-application/debug-service/>

QUESTION 6 / Troubleshooting 30%

Pod cannot resolve `kubernetes.default`. Which two steps are appropriate to check DNS in the cluster?

- A. Check /etc/resolv.conf inside the test Pod
- B. Check Service and log of CoreDNS in kube-system
- C. Delete all NetworkPolicy in all namespaces
- D. Change Service to LoadBalancer

Answer: A, B - Verified

The resolver configuration in the Pod indicates the nameserver/search domain in use; Service, EndpointSlice, and CoreDNS logs confirm cluster-side query paths.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/administer-cluster/dns-debugging-resolution/>

QUESTION 7 / Troubleshooting 30%

`kubectl top nodes` reports Metrics API not available. Which components need to be checked first?

- A. Metrics Server and APIService metrics.k8s.io
- B. Default IngressClass
- C. Default StorageClass
- D. Helm repository cache

Answer: A - Verified

`kubectl top` reads Resource Metrics API. Metrics Server usually provides this API and APIService must be in Available state.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/debug/debug-cluster/resource-metrics-pipeline/>

QUESTION 8 / Troubleshooting 30%

Pod reports ImagePullBackOff with private image. Which two pieces of data need to be verified first?

- A. Image name/tag in Pod spec
- B. Pod events and imagePullSecrets
- C. Reclaim PV's policy
- D. CoreDNS replica number

Answer: A, B - Verified

Wrong image/tag or missing registry credential are common causes. Events often specify not found, unauthorized or timeout errors.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/debug/debug-application/debug-pods/>

QUESTION 9 / Troubleshooting 30%

Which command lists Events by creation time to observe the most recent error sequence?

```
kubectl get events -A --sort-by=.metadata.creationTimestamp
```

- A. The above command is appropriate
- B. kubectl explain events --recursive
- C. kubectl top events -A
- D. kubectl logs events --all

Answer: A - Verified

`--sort-by` sorts objects by the JSONPath field. Viewing Events over time helps connect symptoms to the most recent changes.

Source: <https://v1-35.docs.kubernetes.io/docs/reference/kubernetes-api/cluster-resources/event-v1/>

QUESTION 10 / Troubleshooting 30%

Pod is still running but does not receive traffic and restarts gradually. Which two observations help distinguish probes?

- A. Readiness probe errors will remove the Pod from the backend Service
- B. Liveness probe recurring errors can cause kubelet to restart the container
- C. Readiness probe error always deletes Deployment
- D. Liveness probe only affects DNS records

Answer: A, B - Verified

Readiness controls whether the Pod is ready to receive traffic or not; liveness determines whether the container needs to be restarted to recover.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/configure-pod-container/configure-liveness-readiness-startup-probes/>

QUESTION 11 / Troubleshooting 30%

Control plane kubeadm shows signs of TLS error. Which command checks the certificate expiration date managed by kubeadm?

- A. kubeadm certs check-expiration
- B. kubectl top certificates
- C. kubelet certs renew --all
- D. etcdctl certificate list

Answer: A - Verified

`kubeadm certs check-expiration` displays the expiry date of the PKI managed by kubeadm and helps determine which certificates need to be renewed.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/administer-cluster/kubeadm/kubeadm-certs/>

QUESTION 12 / Troubleshooting 30%

The API server responded but the cluster behaved abnormally. Which endpoint provides details for each readiness check?

- A. kubectl get --raw=/readyz?verbose
- B. kubectl get --raw=/metrics/top
- C. kubectl describe --raw=/api
- D. kubectl logs --raw=/health

Answer: A - Verified

`/readyz?verbose` returns detailed results of readiness checks, useful when the API server is running but a dependency is not ready.

Source: <https://v1-35.docs.kubernetes.io/docs/reference/using-api/health-checks/>

QUESTION 13 / Cluster Architecture 25%

Prepare to maintain a worker node. What two steps create a common safety procedure?

- A. `kubectcl cordon worker-1`
- B. `kubectcl drain worker-1 --ignore-daemonsets`
- C. `kubectcl delete node worker-1 --force`
- D. Delete the entire Pod with `--all-namespaces`

Answer: A, B - Verified

Cordon prevents new workloads from being scheduled; drain uses the Eviction API to move evictable workloads and respects the PodDisruptionBudget.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/administer-cluster/safely-drain-node/>

QUESTION 14 / Cluster Architecture 25%

Which command checks whether user `dev` is allowed to create Deployments in namespace `team-a`?

```
kubectcl auth can-i create deployments --as=dev -n team-a
```

- A. The above command is appropriate
- B. Must use `kubectcl explain`
- C. Must log in directly with the dev user
- D. Can only be checked via audit log

Answer: A - Verified

`kubectcl auth can-i` performs authorization check; `--as` simulates identity if the runner has impersonate permission.

Source: <https://v1-35.docs.kubernetes.io/docs/reference/access-authn-authz/authorization/>

QUESTION 15 / Cluster Architecture 25%

When upgrading a multi-node kubeadm cluster, what is the correct general order?

- A. Raise the control plane first, then the worker nodes one by one
- B. Raise all workers simultaneously in front of the control plane
- C. Only upgrade `kubectcl` on personal computers
- D. Delete the cluster then restore Deployment

Answer: A - Verified

The kubeadm process upgrades the control plane first and then upgrades kubelet/kubectcl on the remaining nodes, while complying with the supported version skew.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/administer-cluster/kubeadm/kubeadm-upgrade/>

QUESTION 16 / Cluster Architecture 25%

What data does the control plane's etcd backup protect and what else is needed for a safe restore?

- A. Save the state of the Kubernetes object in etcd
- B. Keep the certificate/key and corresponding endpoint parameters
- C. Automatically save container images in the registry
- D. Completely replace backup manifest/config outside the cluster

Answer: A, B - Verified

Snapshot etcd saves API state. Restore requires the right tools, endpoints and TLS material; Assets other than etcd still need their own backup strategy.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/administer-cluster/configure-upgrade-etcd/>

QUESTION 17 / Cluster Architecture 25%

In kubeadm HA, where should `controlPlaneEndpoint` point to?

- A. Stable endpoint of load balancer in front of API servers
- B. ClusterIP of an application Service
- C. First IP Pod CoreDNS
- D. NodePort of kube-proxy

Answer: A - Verified

A stable endpoint helps clients and nodes connect to a healthy API server when a control-plane node is not available.

Source: <https://v1-35.docs.kubernetes.io/docs/setup/production-environment/tools/kubeadm/high-availability/>

QUESTION 18 / Cluster Architecture 25%

A folder with `kustomization.yaml`. Which kubectl command renders and applies that content directly?

- A. kubectl apply -k ./overlays/prod
- B. kubectl apply -f kustomization.yaml --raw
- C. kubectl rollout kustomize ./overlays/prod
- D. kubectl create chart ./overlays/prod

Answer: A - Verified

`kubectl apply -k` uses integrated Kustomize to build resources from base/overlay and then apply the results to the cluster.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/manage-kubernetes-objects/kustomization/>

QUESTION 19 / Cluster Architecture 25%

After creating a CustomResourceDefinition, what needs to be added to automatically perform business behavior for the custom resource?

- A. A controller/operator monitors and reconciles the resource
- B. A Service of type LoadBalancer is required
- C. A new default StorageClass
- D. Change CoreDNS to DaemonSet

Answer: A - Verified

CRD extends the API and saves custom objects. The new controller/operator is the component that observes the desired state and performs reconciliation.

Source: <https://v1-35.docs.kubernetes.io/docs/concepts/extend-kubernetes/api-extension/custom-resources/>

QUESTION 20 / Cluster Architecture 25%

Which three interface mappings are correct in the Kubernetes architecture?

- A. CNI — networking
- B. CSI — storage
- C. CRI — container runtime
- D. RBAC — container image format

Answer: A, B, C - Verified

CNI, CSI and CRI create standard boundaries for network plugins, storage drivers and container runtimes. RBAC is an authorization mechanism.

Source: <https://v1-35.docs.kubernetes.io/docs/concepts/architecture/>

QUESTION 21 / Cluster Architecture 25%

If you want to install a release if it doesn't exist yet and upgrade it if it already exists, which Helm command is suitable?

- A. `helm upgrade --install web ./chart`
- B. `helm rollback --delete web ./chart`
- C. `helm repo apply web ./chart`
- D. `kubectl helm install web ./chart`

Answer: A - Verified

`helm upgrade --install` uses an idempotent command for both initial installation and subsequent release updates.

Source: https://helm.sh/docs/helm/helm_upgrade/

QUESTION 22 / Cluster Architecture 25%

Join old expired token. Which command creates a new token and prints the full worker join command?

- A. kubeadm token create --print-join-command
- B. kubectl create token node --join
- C. kubelet join --renew-token
- D. kubeadm init phase worker-token

Answer: A - Verified

The command to create a new bootstrap token and print command contains the API endpoint, token and CA certificate hash so workers can join the cluster.

Source: <https://v1-35.docs.kubernetes.io/docs/reference/setup-tools/kubeadm/kubeadm-token/>

QUESTION 23 / Services & Networking 20%

NetworkPolicy has `podSelector: {}` and only declares `policyTypes: [Ingress]` but has no ingress rule, what will it do?

- A. Isolate the ingress of every Pod in the applied namespace
- B. Block egress of every namespace in the cluster
- C. Delete Service EndpointSlice
- D. Only block traffic from the control plane node

Answer: A - Verified

Empty selector selects all Pods in the same namespace. When a Pod is selected for Ingress without a rule allowing it, ingress to the Pod is denied.

Source: <https://v1-35.docs.kubernetes.io/docs/concepts/services-networking/network-policies/>

QUESTION 24 / Services & Networking 20%

Which two descriptions are correct for Service type NodePort?

- A. Still have ClusterIP for internal cluster access
- B. Open the same nodePort on the matching nodes
- C. Create your own Ingress Controller
- D. Required volume level for each backend Pod

Answer: A, B - Verified

NodePort builds on Service ClusterIP and publishes a port on the node. It does not provide HTTP routing itself like Ingress/Gateway.

Source: <https://v1-35.docs.kubernetes.io/docs/concepts/services-networking/service/#publishing-services-service-types>

QUESTION 25 / Services & Networking 20%

Setting `clusterIP: None` for a Service mainly creates what behavior?

- A. Do not grant virtual IP; DNS can return the backend address directly
- B. Self-powered external load balancer
- C. Encrypt all traffic with mTLS
- D. Convert Service to ExternalName

Answer: A - Verified

Headless Service bypasses load-balancing via ClusterIP and allows the discovery mechanism to return the appropriate endpoint/Pod directly.

Source: <https://v1-35.docs.kubernetes.io/docs/concepts/services-networking/service/#headless-services>

QUESTION 26 / Services & Networking 20%

In Gateway API, HTTPRoute is attached to Gateway with what field?

- A. spec.parentRefs
- B. spec.clusterIP
- C. metadata.ownerReferences specifies the Pod
- D. status.loadBalancerClass

Answer: A - Verified

`parentRefs` describes the parent resource receiving the route, usually the Gateway. The Gateway's listener and policy decide whether the route is attached or not.

Source: <https://v1-35.docs.kubernetes.io/docs/concepts/services-networking/gateway/>

QUESTION 27 / Services & Networking 20%

Ingress resource created but traffic is not processed. What foundational conditions are often missing?

- A. Ingress controller matches IngressClass
- B. A PersistentVolume of type hostPath
- C. A new kubeadm token
- D. A static Pod runs on every worker

Answer: A - Verified

Ingress resource only describes the rule. The controller deploys the dataplane and observes the new IngressClass to turn the rule into actual routing.

Source: <https://v1-35.docs.kubernetes.io/docs/concepts/services-networking/ingress-controllers/>

QUESTION 28 / Services & Networking 20%

In most clusters, which command is useful to check the system Service DNS?

- A. `kubectl -n kube-system get svc kube-dns`
- B. `kubectl get ingress coredns -A`
- C. `kubectl top storageclass kube-dns`
- D. `kubectl drain coredns`

Answer: A - Verified

Even if the DNS implementation is CoreDNS, the Service is usually named `kube-dns` to maintain compatibility.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/administer-cluster/dns-debugging-resolution/>

QUESTION 29 / Services & Networking 20%

After `kubeadm init`, CoreDNS is still Pending and the node is not Ready. Which components usually need to be installed next?

- A. A compatible CNI network add-on
- B. A default Ingress resource
- C. A PersistentVolume for kubelet
- D. An HPA for CoreDNS

Answer: A - Verified

kubeadm không cài Pod network. CNI add-on cung cấp networking cần thiết; CoreDNS thường chỉ schedule bình thường sau khi network được cấu hình.

Source: <https://v1-35.docs.kubernetes.io/docs/setup/production-environment/tools/kubeadm/create-cluster-kubeadm/>

QUESTION 30 / Services & Networking 20%

Setting `externalTrafficPolicy: Local` for Service NodePort/LoadBalancer typically creates what two effects?

- A. Can keep the source IP address of the client
- B. Node only forwards traffic to the appropriate local endpoint
- C. Self-encrypt traffic with TLS
- D. Completely bypass kube-proxy or equivalent dataplane

Answer: A, B - Verified

The Local policy avoids jumping to an endpoint at another node, thereby preserving the source IP; In return, traffic to a node without a local endpoint may not be served.

Source: <https://v1-35.docs.kubernetes.io/docs/tutorials/services/source-ip/>

QUESTION 31 / Workloads & Scheduling 15%

Deployment just rolled out image error. Which command returns the workload to revision first?

- A. `kubectl rollout restart deployment/web`
- B. `kubectl rollout undo deployment/web`
- C. `kubectl scale deployment/web --replicas=0`
- D. `kubectl replace deployment/web`

Answer: B - Verified

``kubectl rollout undo`` returns to the previous revision of Deployment. You can add ``--to-revision`` when you need to select a specific revision.

Source: <https://v1-35.docs.kubernetes.io/docs/concepts/workloads/controllers/deployment/#rolling-back-a-deployment>

QUESTION 32 / Workloads & Scheduling 15%

Pod gets ConfigMap via environment variable. After ConfigMap changes, how does the process receive the new value?

- A. Recreate/rollout the Pod so that the environment variables are reloaded
- B. No need to do anything because the env will change immediately
- C. Just restart kube-proxy
- D. Change ConfigMap to StorageClass

Answer: A - Verified

Environment variables are set when the container starts and do not update themselves. Volume from ConfigMap has a different update mechanism, but the application still has to re-read the file.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/configure-pod-container/configure-pod-configmap/>

QUESTION 33 / Workloads & Scheduling 15%

What two conditions does HPA by CPU need to calculate a stable resource utilization rate?

- A. Has a working Resource Metrics API
- B. The container declares appropriate CPU requests
- C. Service must be LoadBalancer
- D. Pod is required to use hostNetwork

Answer: A, B - Verified

HPA needs current metrics and CPU requests as denominators for utilization. Missing requests can cause the Pod's metrics to not be used in calculations.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/run-application/horizontal-pod-autoscale/>

QUESTION 34 / Workloads & Scheduling 15%

If you want the Pod to only be scheduled to the node with label `disk=ssd`, what type of node affinity represents the required requirement?

- A. `requiredDuringSchedulingIgnoredDuringExecution`
- B. `preferredDuringSchedulingIgnoredDuringExecution`
- C. `podAntiAffinity` with only empty `topologyKey`
- D. toleration without taint

Answer: A - Verified

`requiredDuringSchedulingIgnoredDuringExecution` is a hard condition at schedule time; `preferred...` only increases the priority score for the appropriate node.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/configure-pod-container/assign-pods-nodes-using-node-affinity/>

QUESTION 35 / Workloads & Scheduling 15%

`PodDisruptionBudget` primarily protects workloads against what type of disruption?

- A. Voluntary disruption like drain node
- B. Any unexpected hardware failure
- C. Any code errors in the container
- D. Image registry is deleted

Answer: A - Verified

PDB limits the number of Pods that can be voluntarily interrupted at the same time via the Eviction API. It does not prevent unintended node or application crashes.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/run-application/configure-pdb/>

QUESTION 36 / Workloads & Scheduling 15%

The application starts up very slowly but then stabilizes. Which probe helps delay liveness/readiness until the startup is complete?

- A. `startupProbe`
- B. `executeLifecycleProbe`
- C. `schedulerProbe`
- D. `endpointProbe`

Answer: A - Verified

When `startupProbe` is configured, liveness and readiness do not begin until startup is successful, avoiding mistakenly restarting slow-starting applications.

Source: <https://v1-35.docs.kubernetes.io/docs/tasks/configure-pod-container/configure-liveness-readiness-startup-probes/>

QUESTION 37 / Storage 10%

PVC uses dynamic StorageClass but still Pending. What information should I check first?

- A. Events of PVC and provisioner of StorageClass
- B. Log of kube-proxy
- C. CoreDNS replica number
- D. Image pull policy of Pod

Answer: A - Verified

PVC events usually indicate which provisioner, quota, topology or access mode is causing the volume to not be allocated.

Source: <https://v1-35.docs.kubernetes.io/docs/concepts/storage/persistent-volumes/>

QUESTION 38 / Storage 10%

Dynamic PV has reclaim policy `Retain`. When PVC is deleted, what happens to storage assets?

- A. Assets are retained for manual data retrieval by administrators
- B. Assets are always deleted immediately
- C. PV converts itself to StorageClass
- D. The PVC itself is recreated in the default namespace

Answer: A - Verified

`Retain` keeps the resource stored after the claim is cleared. The PV changes state and requires a manual data recall/delete process before reuse.

Source: <https://v1-35.docs.kubernetes.io/docs/concepts/storage/persistent-volumes/#reclaiming>

QUESTION 39 / Storage 10%

Which access mode ensures that a volume can only be mounted read-write by one Pod in the entire cluster?

- A. ReadWriteOncePod (RWOP)
- B. ReadWriteOnce (RWO)
- C. ReadOnlyMany (ROX)
- D. ReadWriteMany (RWX)

Answer: A - Verified

RWOP is limited to one Pod. RWO limits the read-write volume on a node and multiple Pods on the same node can still access it depending on the driver.

Source: <https://v1-35.docs.kubernetes.io/docs/concepts/storage/persistent-volumes/#access-modes>

QUESTION 40 / Storage 10%

StorageClass uses `volumeBindingMode: WaitForFirstConsumer`. How long is binding/provision volume delayed?

- A. When there is a Pod, use PVC and the scheduler knows the appropriate topology
- B. When the cluster creates the first Service LoadBalancer
- C. When kubelet restarts on every node
- D. When the PVC is removed

Answer: A - Verified

This mode allows the scheduler to consider the Pod's zone/node topology before selecting or provisioning volumes, avoiding creating volumes in locations that cannot be scheduled.

Source: <https://v1-35.docs.kubernetes.io/docs/concepts/storage/storage-classes/#volume-binding-mode>